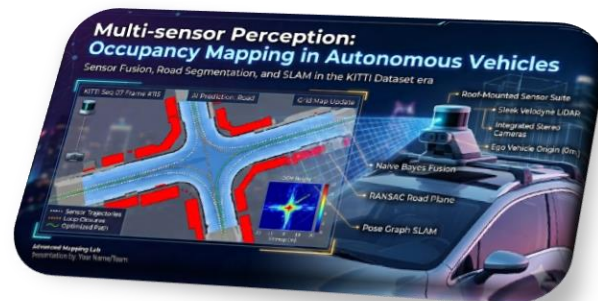




Project 1

Topics:

- Geodetic coordinate system and the KITTI dataset
- Probabilistic Occupancy Grid
- Sensor fusion and semantic segmentation

During this project, you will get experience with recorded data from a mobile lab and practice implementing the algorithm for robot mapping which you learned in the class. It is recommended to use the instructions in the attached Google Colab script, but please copy all the figures to the final report. Analysis of the results is a significant part of the project. To diagnose the details and explanations is what is expected from you.

Notice that each of you is given a different scenario to analyze, therefore your conclusions and results should differ from one to the other.

Data Category	name	Last ID
City	2011_09_26_drive_0104	0-1
City	2011_09_26_drive_0095	2-3
Residential	2011_09_26_drive_0064	4-5
City	2011_09_26_drive_0013	6-7
Residential	2011_09_26_drive_0061	8-9

**** The KITTI vision benchmark suite**

- [Link](#)
- [paper](#)

Part A: Geodetic coordinate system and get familiar with the KITTI dataset (20%)

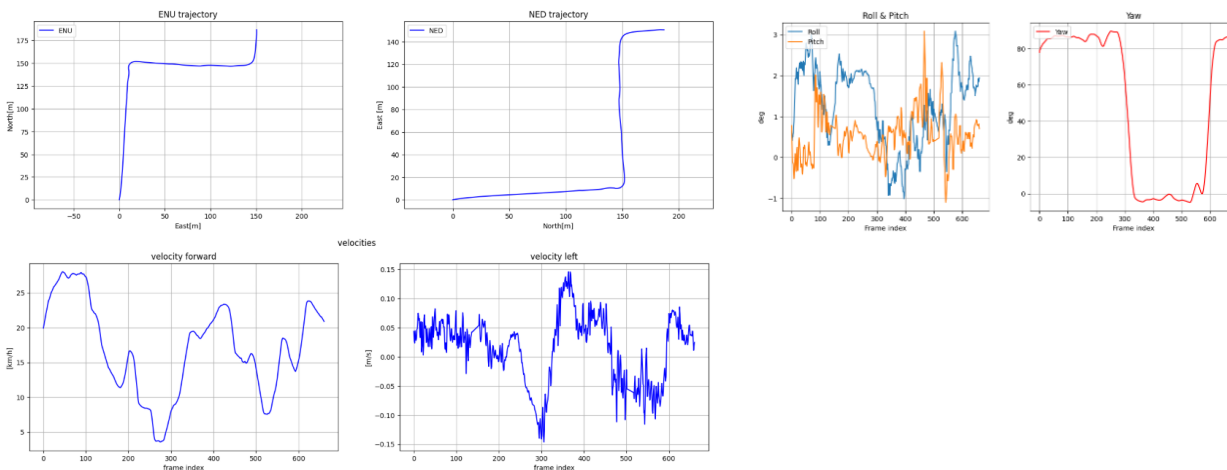
1. Download your recording data according to the table below from the KITTI dataset.
 - a. Describe technical details about your scenario such as driving route, presence of dynamic objects (cyclists, cars and pedestrians), occlusions, turns etc. (3%)
 - b. Use Google Maps to find the trajectory of the recorded data. Plot the results.

Example: (the trajectory of the car is marked in red) (2%)



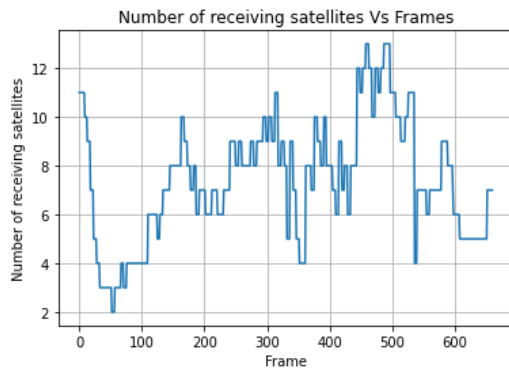
- c. Display the trajectory in local **ENU** and **NED** coordinate frames. Plot the measured velocities (**forward** and **right** components). Also plot the **orientation** (roll, pitch, yaw).

Explain how the vehicle trajectory is related to the measured yaw and velocity components, and describe how turning maneuvers are manifested in the ENU/NED trajectories and in the forward and lateral velocity plots. (5%)



d. Display the quality of GPS signal reception and any relevant details. Explain the causes and consequences of variations in GPS satellite reception in your scenario.

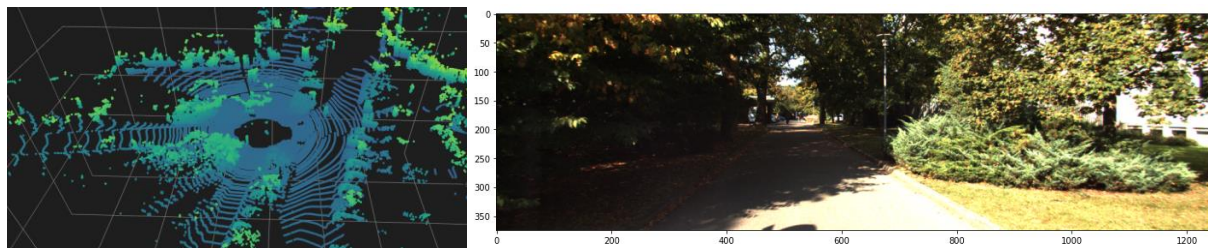
Example: (5%)



e. Present two one example of a camera image and its corresponding registered LiDAR point cloud from the dataset.

Identify elements in the scene that are difficult to perceive using only the camera or only the LiDAR. Explain the limitations of each sensor and how sensor fusion addresses these challenges. (5%)

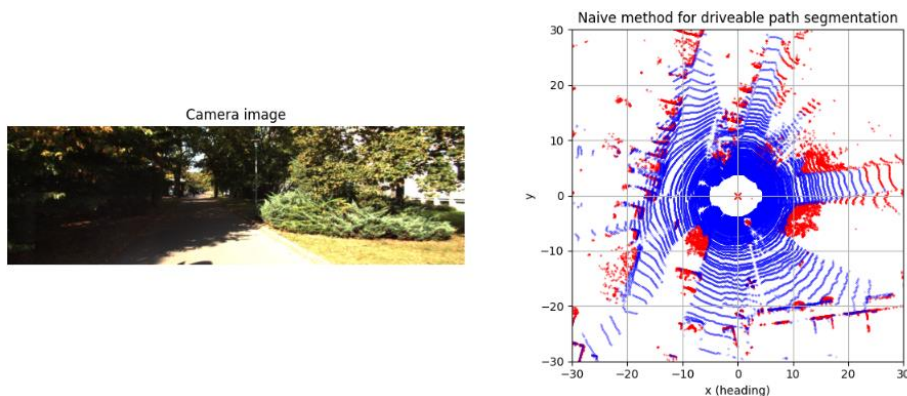
Example:



Part B: Probabilistic Occupancy Grid (35%)

1. Implementation of a grid on a single scan based on the following instructions: (15%)
 - a. To find a drivable path in the session, use a naïve role: filter all the point clouds above ~30 cm off the ground (assume flat surface).

For example,

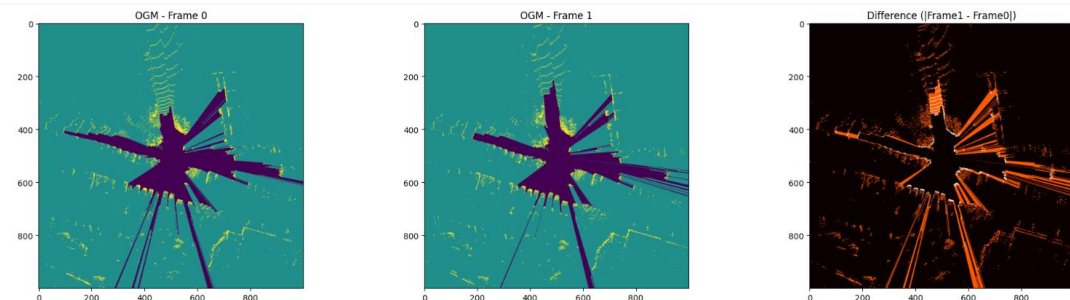


b. Build occupancy map!

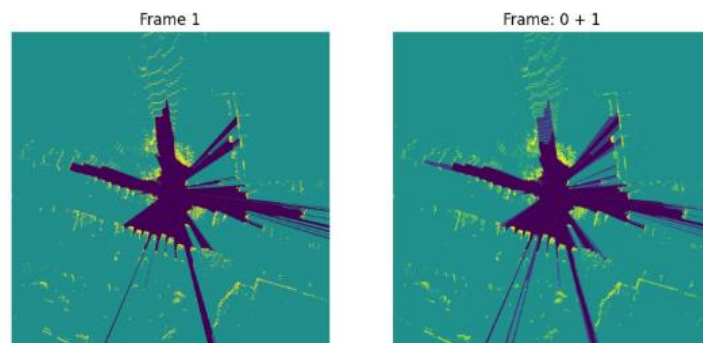
Mode parameter calibration- you can optimize the parameters for the scenario. The recommended hyperparameters are listed below, but they are flexible. Describe your implementation and display result.

Parameter	Value	Description
Resolution	0.2 (m)	cell size
ALPHA	0.2 (m)	The radial resolution when converting LiDAR data to grid map
BETA	0.1 (degree)	The angular resolution when converting LiDAR data to grid map
Max range	100 (m)	Maximum range of LiDAR points that will be converted to grid map.
Hit/Miss prob	0.8/0.2	Probabilities of hit and miss values
Probability Saturation	0.02/0.98	Minimum and maximum probability
Map width	100 (m)	Width of the map from side to side
SPHERICAL2CARTESIAN_BIAS	1 (pixels)	An adjustment needed due to some errors when converting the spherical grid map to the cartesian grid map.

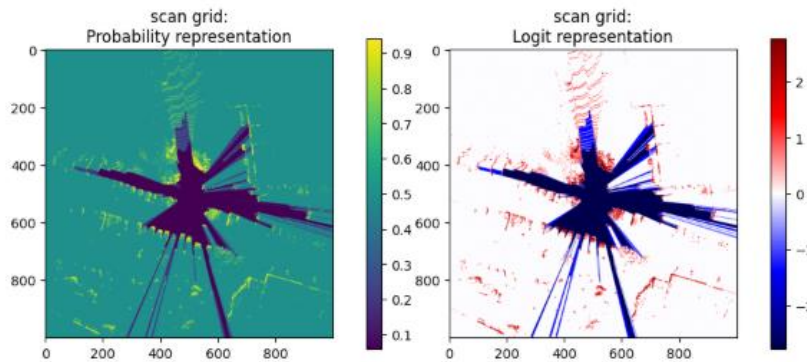
- c. Display the occupancy-grid result for two consecutive frames (e.g., Frame 0 and Frame 1) and also display their difference map (Frame 1 – Frame 0). Describe your implementation, and explain—based on your scenario—what additional information the second frame contributes compared to the first. Also state what you expect the fusion of the two frames to produce. If the first two frames look almost identical, choose a different pair of frames where the change is clearer. Example: Expected results:



- d. Update and shift the OGM (Occupancy Grid Map) according to the next pose of the car. Show an example of the first two frames. Example: Expected results:



e. Show segmentation map based on Logit mechanism.



2. Robot mapping!! Implementation of a Probability Occupancy Grid over all the record: (10%)

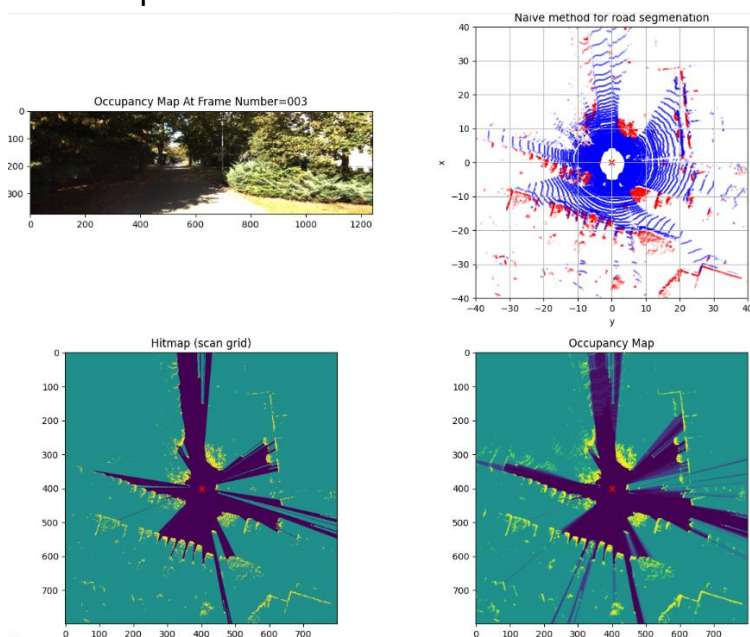
a. Pre-processing-

- i. Transform the point cloud recorded (in previous section) by the LiDAR to INS location.
- ii. Align the point cloud based on the IMU inputs.

b. Build an OGM per LiDAR scan (use the function in section 1)

c. Repeat the occupancy map process over all your scans/frames!

d. Create an animation which includes the following sub plots like in the following example:



e. Describe your work process (explain flow + code).

f. Add to the report the last frame of the occupancy map.

g. Apply thresholds on the OGM (chose one frame from the record) to get segmented map: The segmented maps should describe 3 types of cells: free, occupied and unknown. Add the figures of the first scan: filter point cloud, scan grid in probability and logit representations. Please describe your implementations in detail.

3. Analyze your results. (15%)

- a. Please show examples of good and bad examples (at least two of each) of mapping during the scene and explain what makes each example good or bad.
- b. Change any **hyperparameter**** of the model in a way you choose to improve the results from a specific perspective, either for a particular scene or across all scenes (e.g., dynamic objects, resolution, speed, etc.), and run the model again.
 - i. Describe the change you made, the final results, how it affected the map, and what you expect to observe in the mapping results.
 - ii. Compare and analyze the effect of this change, using examples from interesting cases.
 - iii. Add a new animation based on your selected experimental parameter

***for examples:*

- *Hit/Miss probabilities to 0.6/0.4..*
 - *Pixel size to 0.3..*
 - *Probability saturation values..*
 - *Alfa/beta..*
- c. Describe how objects of different types affect the mapping? For example:
 - Dynamic objects.
 - Parking cars /trees.
 - Sparse/dense objects.
 - Transparent materials (such as glass).
 - Different color objects of the same type..

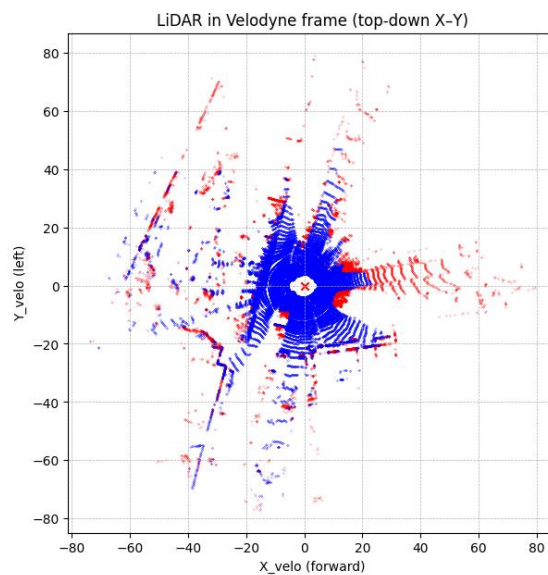
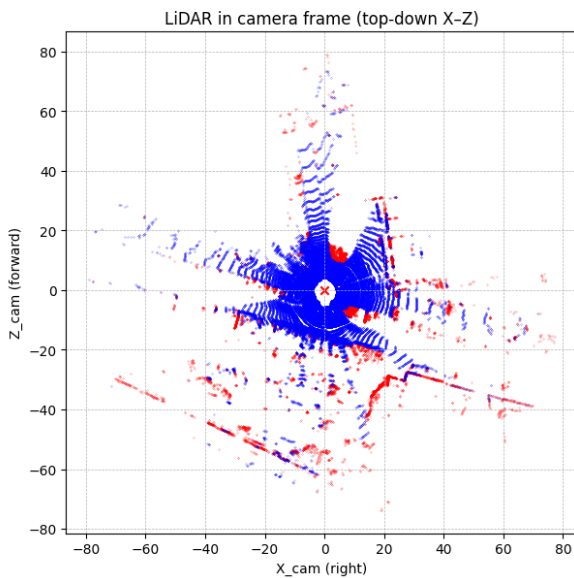
Please explain and show examples from your record.

Part C: Sensor fusion and semantic segmentation (40%)

1. **Display obstacle vs road using camera coordinates** -The goal of this section is to classify LiDAR points into **road** and **obstacle** classes based on height. (5%)

You are required to complete and run code that:

- (a) filters the point cloud by removing points below the sensor and points close to the vehicle
- (b) transforms the LiDAR points from the Velodyne coordinate system to the camera coordinate system using the extrinsic matrix
- (c) performs height-based segmentation and visualizes the result in a top-down view (red = obstacle, blue = drivable path).
- (d) transform the points back to the Velodyne frame and compare the segmentation results in both coordinate systems using different height thresholds.

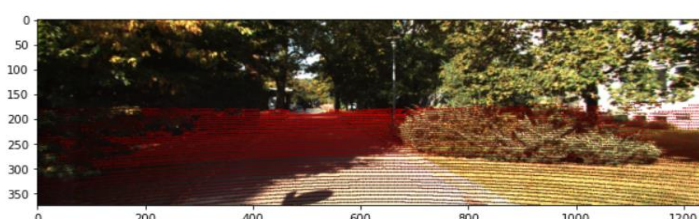


2. **Sensor fusion: Mapping LiDAR point clouds to camera image coordinates as follows:** (5%)

- Remove point cloud data located behind the camera.
- Transform and project the remaining points onto the image plane.
- Discard projected points that fall outside the image boundaries.

- a. Describe the working procedure (explain the processing pipeline).
- b. Include 3 examples of the projection results onto image coordinates in the report

Example:

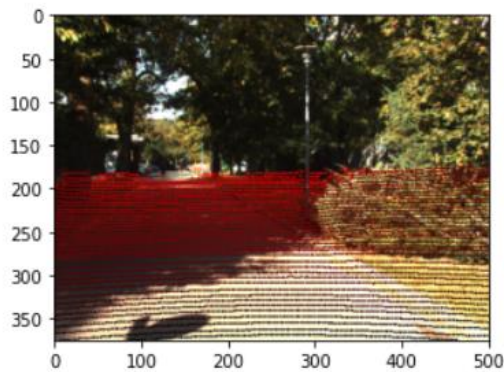


3. Road segmentation! (5%)

- a. Pre-processing: Image cropping- The deep learning model (DeepLabv3+) was trained with cropped images from the KITTI dataset with aspect ratio 4:3 (W:H), which was resized further to 513 x 513 images.

crop the image (from the previous section) from the center and add it to your report.

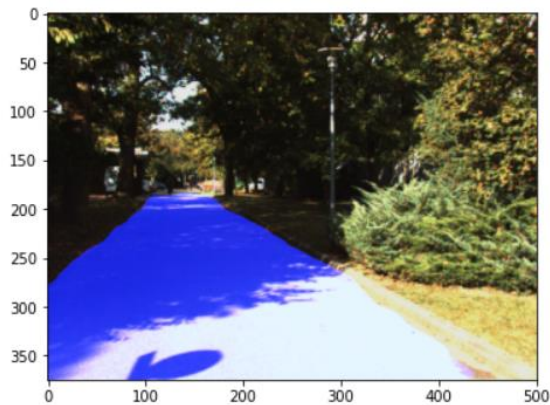
Example:



- b. Run DeepLabV3+ (use the pretrained model) on your image.

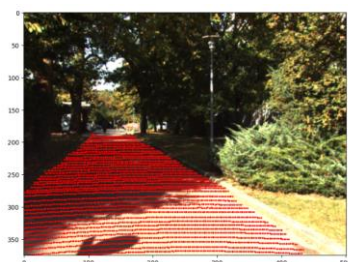
Describe the process of using the algorithm and attach the results to the report:

Example: (road is marked in blue)



- c. Extract from the point cloud only the points that correspond to the road.
Describe your process and attach the figure to the report:

Example:



4. LiDAR Road Filter (5%)

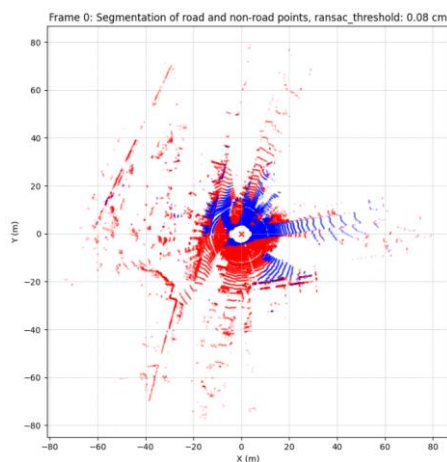
The road points set that are obtained from the previous step may contain false detections and obviously only contains points that are inside the camera field of view. We also need to detect road points that are outside this set!

To do this, fit a plane model (which will represent the road model) $Ax+By+Cz=1$ to the current road points. Any points that are outside the camera's field of view and located near the road plane model can also be regarded as road points. RANSAC algorithm is applied when fitting the model to reduce the influence of outliers.

Parameter for RANSAC model:

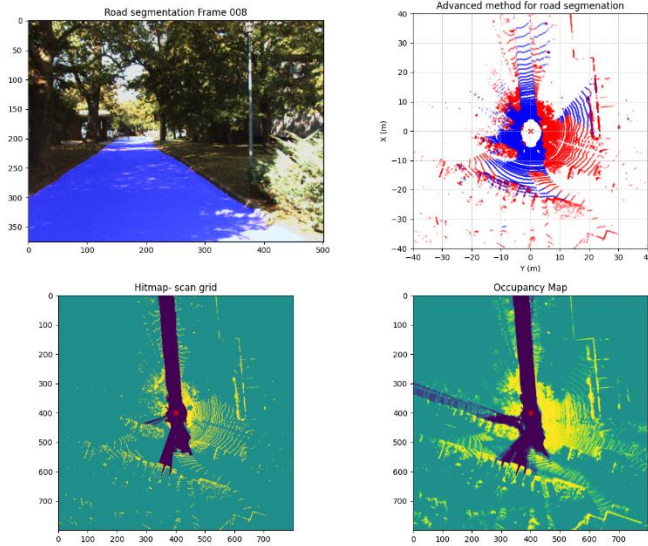
- ROAD_HEIGHT_THRESHOLD - The maximum distance (in height axis) between a point to the road model to be considered as road point
- a. Calculate the plane model from the road points.
 - b. Predict the drivable path across all LiDAR point clouds using the plane model estimated in the previous section. Compare and analyze the lower threshold versus the threshold used in the naive method.

Example: In blue — the predicted path based on DeepLabv3+ and the projection mechanism using a threshold of $TH = 0.08$ cm



5. Robot mapping based on AI! Implementation of a probability occupancy grid based on deep learning! (10%)

- a. Repeat the occupancy map process based on deep learning over all your scans/frames!
- b. Create an animation which include the following sub plots like in the following example:



- c. Describe your work process (explain flow + code).
- d. Add the animation to the report.

6. Analyze your results. (10%)

- a. Compare and analyze the new mapping based on deep learning to the naïve method (section B).
- b. Please show examples of good and bad examples (at least two of each) of mapping with deep learning during the scene and explain what makes each example good or bad.
- c. Describe at least three of the pros and cons to mapping based on deep learning vs the naïve method.
- d. Mention at least 2 algorithmic challenges in road segmentation for offroad.

The final report should include the following details:

- Analysis and explanations (Hebrew or English)- explain in detail and use examples to support your descriptions.
- Each graph/animation/figure must include titles, units, and descriptions.
- Attach the code in addition to your report.
- Attach the animations (AVI or MP4 format). Keep frame rate to a maximum of 10Hz.
- The code should be a script that displays and plots all the results. Verify that the code works properly and includes all the sub-functions. In case of a code which does not run/work properly/fail you might lose points.

Good Luck!

Appendix

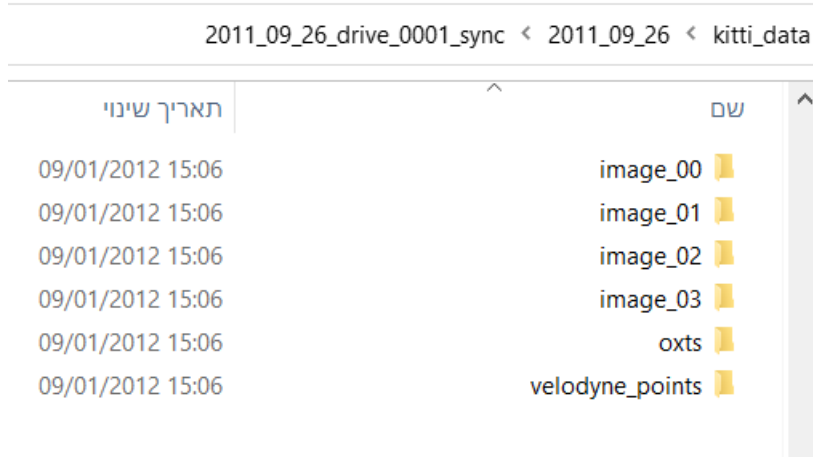
- Please be honest, you may automatically lose points if you are caught copying including from the internet (code, results). The work is personal.
- See instructions about the recorded data in the Appendix.
- You are required to read the following paper for better understanding.
 - Vision meets Robotics: The KITTI Dataset/ Andreas Geiger
- SW – Python
- The final grade is given according to the quality of your analyses, descriptions, conclusions, explanations, the form of the results (plot, graphs, movies), understandable code with comments and explanations. It is possible that the final performance and results will not be as perfect as you desired, as this is real data and is part of the challenge of the autonomous driving field. Feel free to suggest solutions that could improve your results if this is the ca

A. The KITTI vision benchmark suite

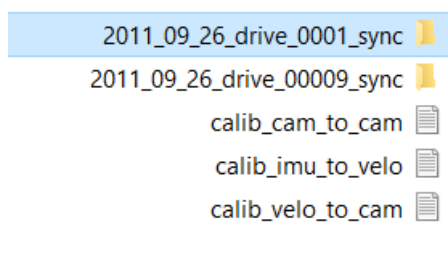
1. [Link](#)
2. Read the [paper](#)
3. Download the [raw data development kit \(1 MB\)](#)
4. Download your data according to the below table.
□download [\[synced+rectified data\]](#) [\[calibration\]](#)
5. Keep the format:
The data format saved in <Folder of date>\<Title>
For example:
[2011_09_26](#)\[2011_09_26_drive_0009_sync](#)

6. Keep the following format:

- You must save all the data in the same directory name as directory name



- Save calibration files (transformation) in <Folder of date>, as follow:



7. Recorded Data (Go to raw data). (download [\[synced+rectified data\]](#) [\[calibration\]](#)).



B. Your final package should contain the following folders:

- **Report** contains the report
- **Code** contains all functions + sub-functions
- **Results** stores the resulting figures ,movies, etc.

Save the package as <name>_<ID>.zip

In moodle upload:

Save the package as <name>_<ID>.zip

Report_<name>_<ID>.pdf